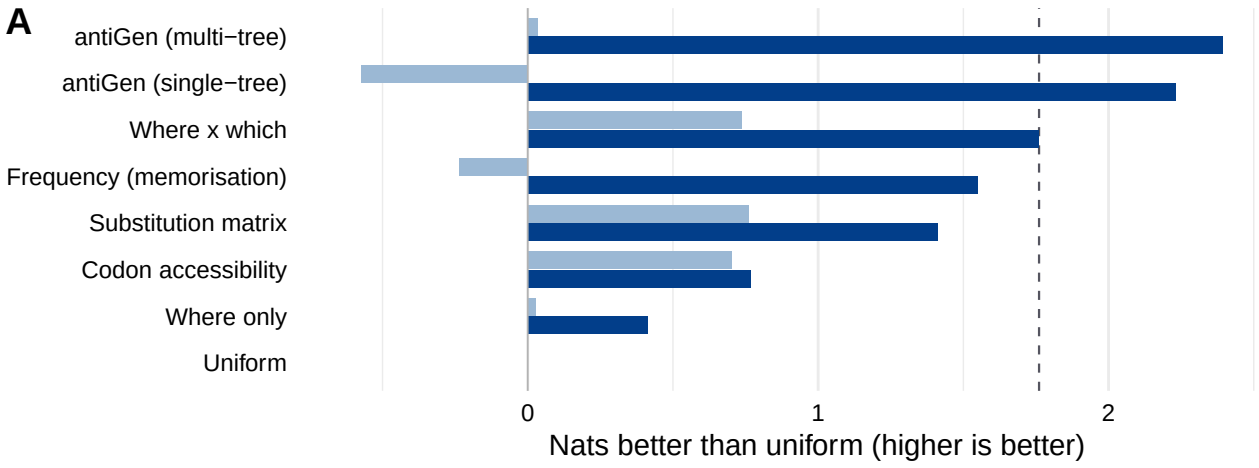
All substitutions
Novel only

Random holdout
held−out nodes, same trees

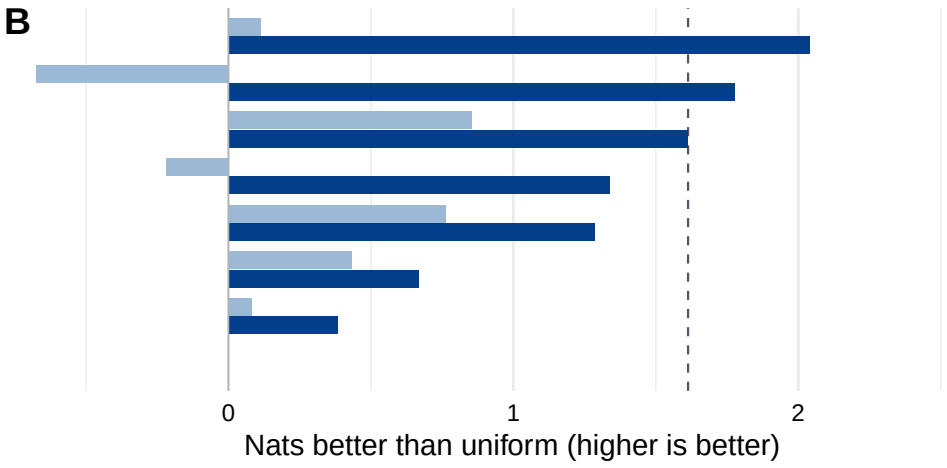
Temporal holdout
D = 2025−01−01, unseen future

Ladder: NLL

A

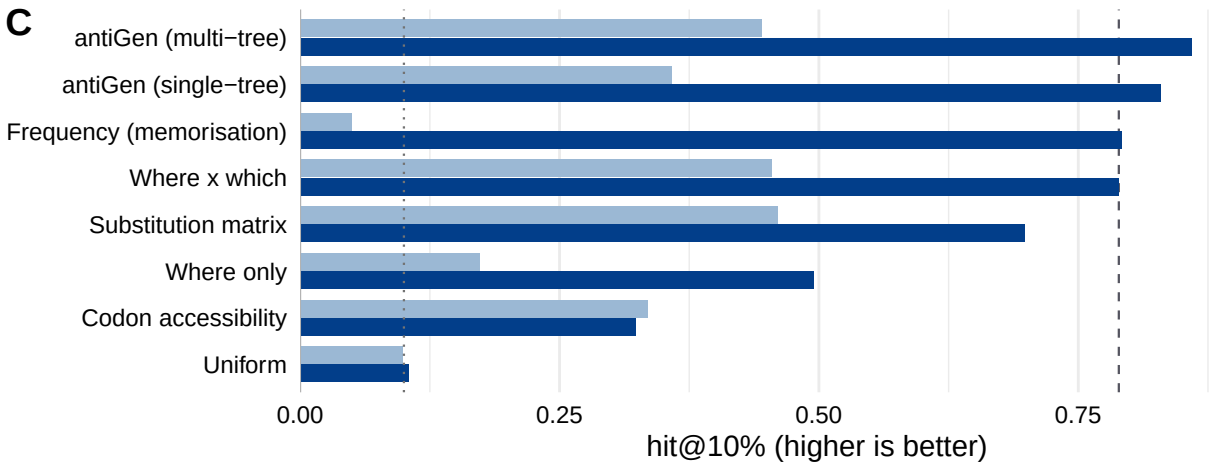


B

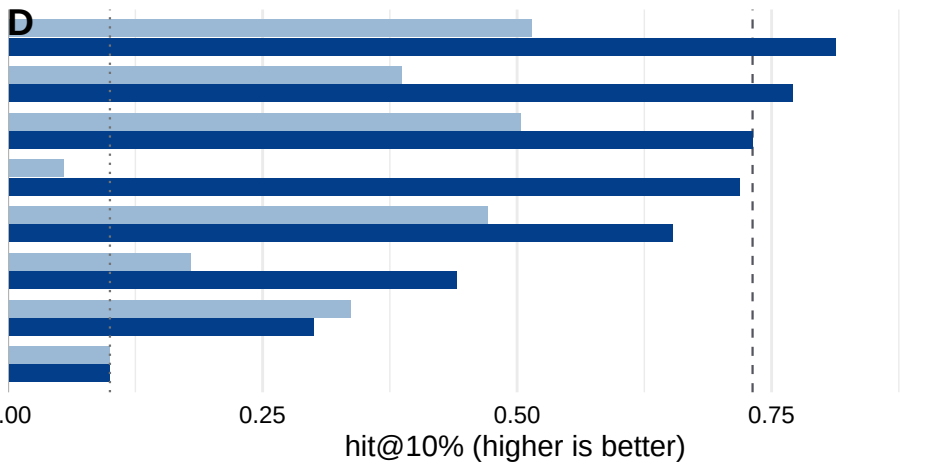


Ladder: hit@10%

C

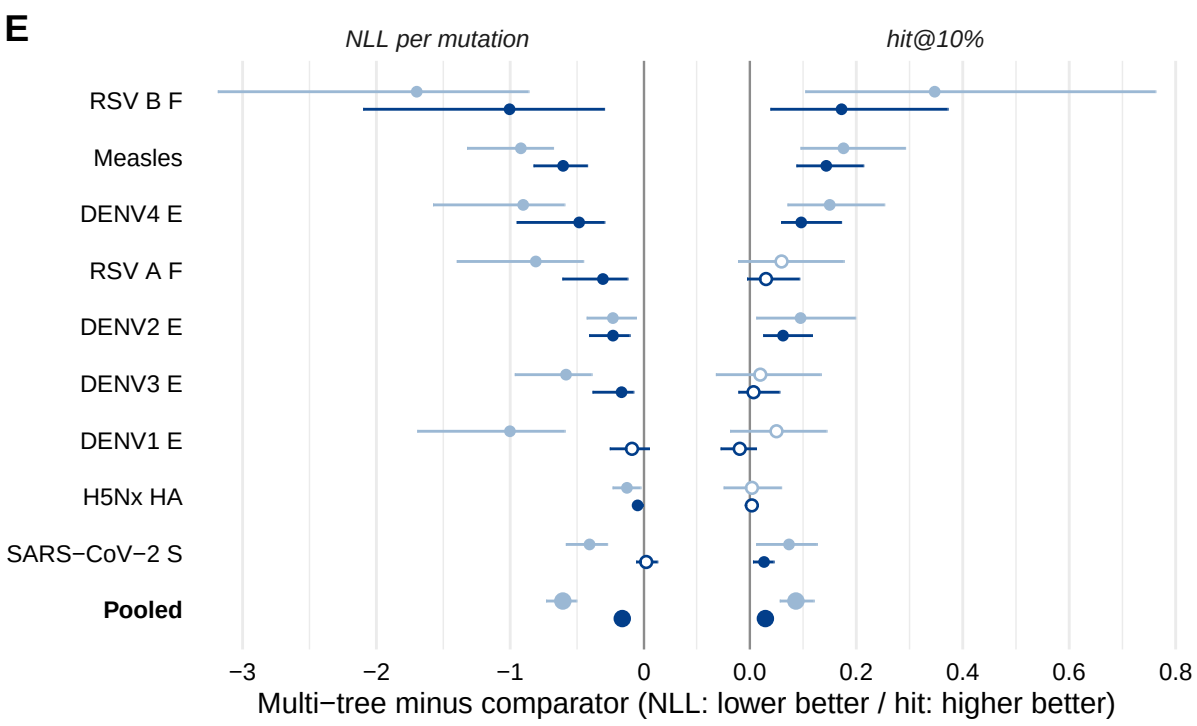


D

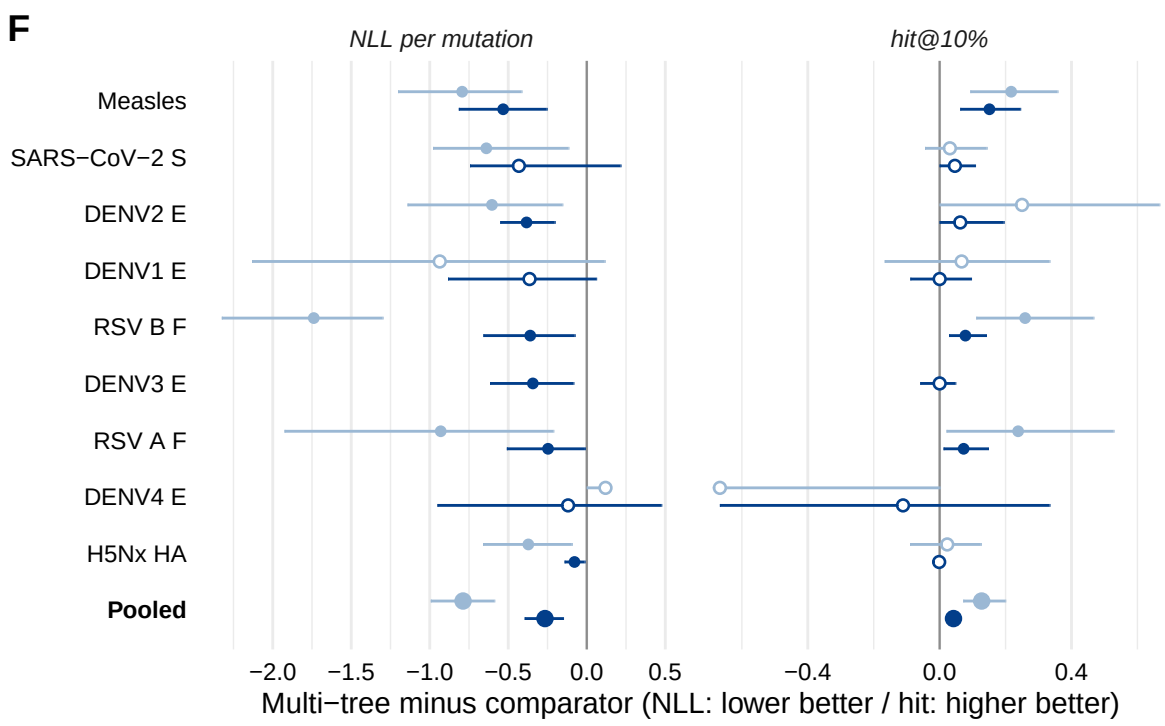


vs single−tree

E

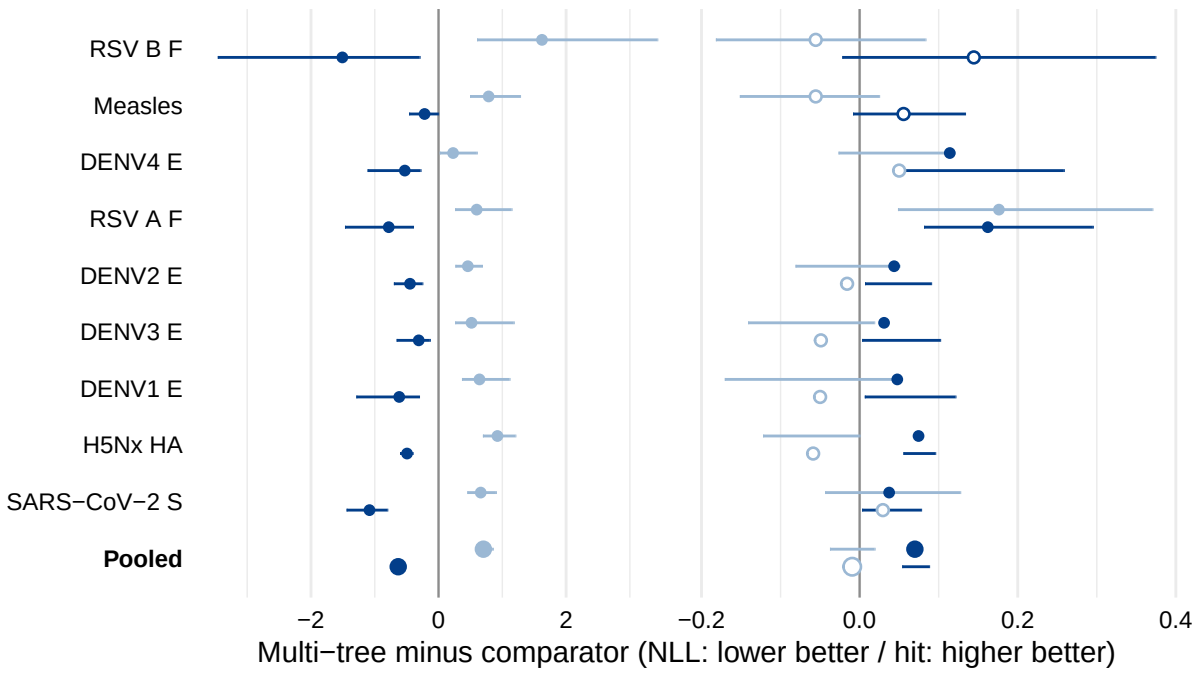


F

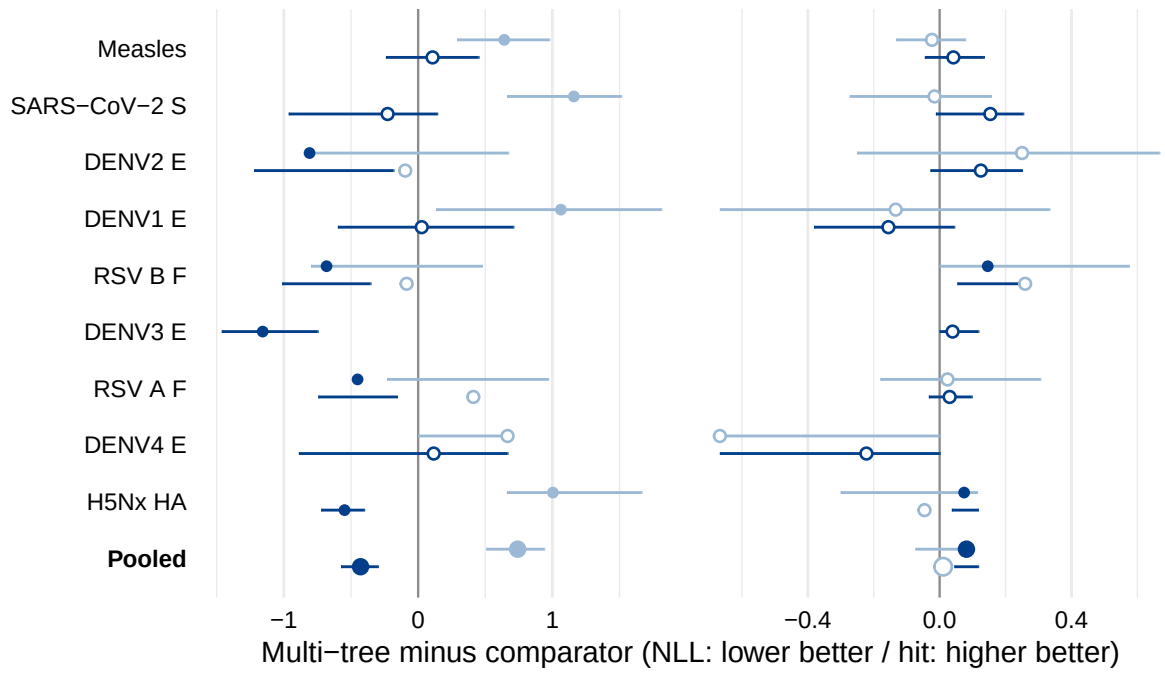


vs where x which

G



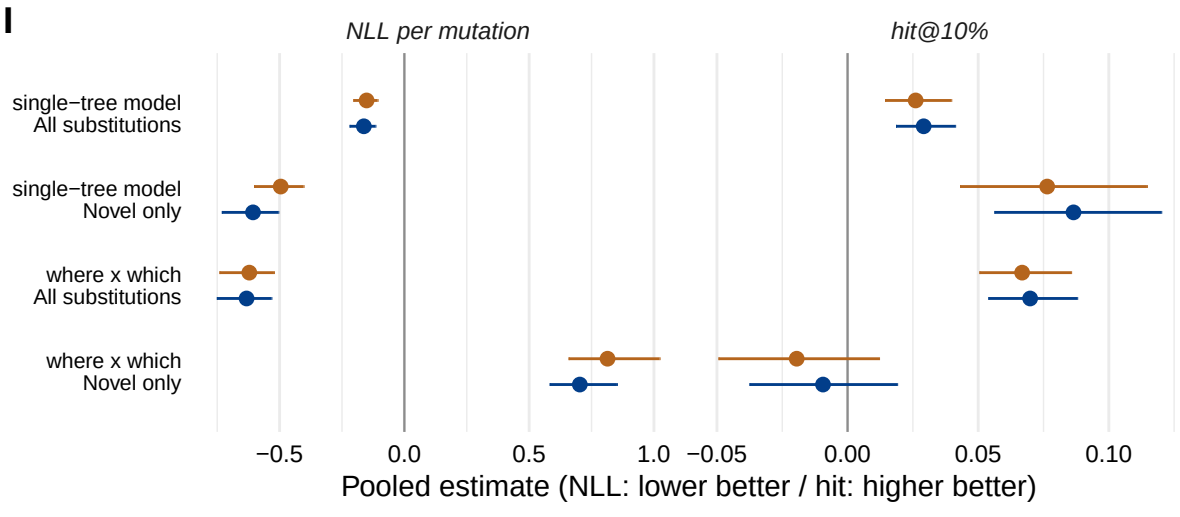
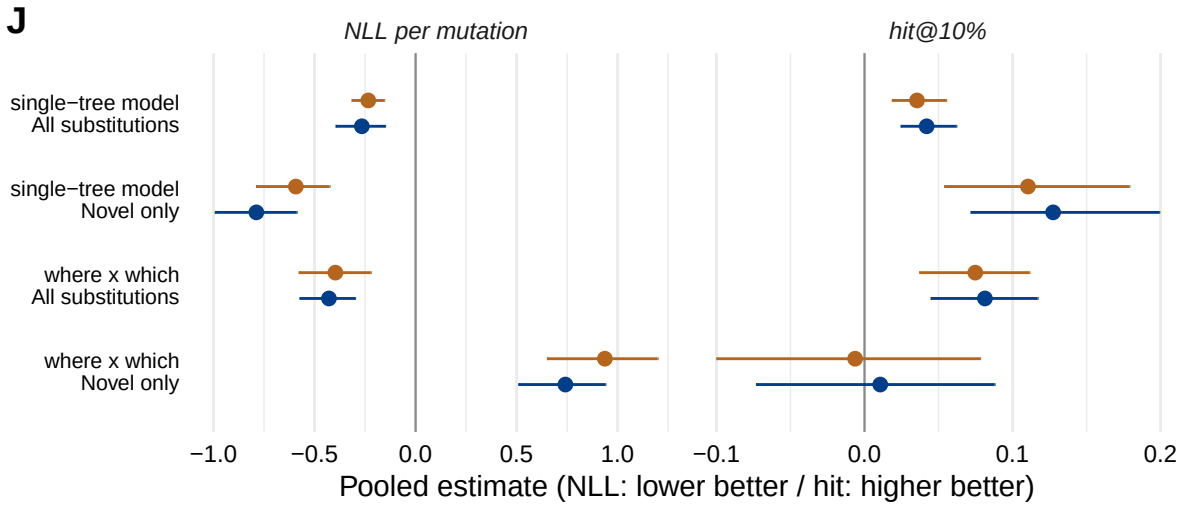
H



Pooled−validation epoch (deployed)
Per−tree best epoch

Epoch rule

I



J